

Research Statement

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My research lies at the interface of analysis, geometry, topology, and mathematical physics. A central question guiding my work is: *how can techniques from global analysis and index theory be used to understand spectral problems and to construct and compare geometric and topological invariants?* A recurring theme in my work is to convert global or singular problems into local geometric analysis. Witten deformation concentrates spectral information near critical sets, heat-kernel analysis connects local geometry with global spectral invariants, and compactifications of configuration spaces provide a geometric framework for analyzing the singularities in Feynman graph integrals. My research develops along three interconnected directions:

- (a) Feynman graph integrals in holomorphic quantum field theory and their applications to higher-genus B-model invariants;
- (b) analytic torsion and higher Cheeger–Müller theory, using Witten deformation to compare analytic and topological invariants;
- (c) spectral theory of Schrödinger-type operators on noncompact manifolds, motivated in part by Landau–Ginzburg models.

These directions come together in my long-term program toward a higher-genus Calabi–Yau/Landau–Ginzburg correspondence on the B-model side.

1 Overview of my results

Feynman graph integrals and holomorphic field theory. With Minghao Wang, we proved the convergence of Feynman graph integrals on closed real-analytic Kähler manifolds in the sense of Cauchy principal value [22]. On Calabi–Yau threefolds, this gives a direct and rigorous realization of the candidate higher-genus B-model invariants proposed by Bershadsky–Cecotti–Ooguri–Vafa. We subsequently proved that three natural rigorous renormalization procedures give the same answer and derived a gauge anomaly formula [23]. Such agreement is not generally expected in QFT, where different renormalization schemes may produce different finite parts.

Analytic torsion and higher torsion. Using Witten deformation by specially chosen non-Morse functions, I developed a localization method adapted to decompositions of manifolds. This gives a new purely analytic proof of the gluing formula for analytic torsion and, for families, a simpler proof of the gluing formula for Bismut–Lott torsion forms [24, 25]. With Martin Puchol [19], we proved a comparison of analytic and topological higher torsions for unitary flat bundles using generalized Morse functions. This removes the fiberwise Morse-function existence assumption in [3] and trivial flat bundle assumption in [20], while retaining Thom–Smale transversality. The proof resolves the main analytic difficulty caused by birth–death singularities.

Spectral theory on noncompact manifolds. With Maxim Braverman and Xianzhe Dai [8], we obtained an intrinsic criterion for the classical Weyl law for Schrödinger operators on general complete manifolds, removing both bounded geometry and the doubling condition from our earlier result [12]. We also constructed examples showing that the Weyl law can fail when the criterion is violated [8]. More recently, I proved that the weighted Laplacian on every complete gradient Ricci shrinker satisfies the classical Weyl law [26]. My earlier work with Dai established Agmon estimates, a comparison of the analytic Witten complex with a finite-dimensional complex generated by critical points, heat-kernel expansions and a local index theorem [10, 11].

2 Feynman graph integrals and holomorphic field theory

In perturbative quantum field theory, Feynman graph integrals encode the contributions associated with interactions of particles. In mathematics, they also provide a striking bridge between physical theories and geometric or topological invariants. In topological quantum field theory (TQFT), such integrals converge in the Lebesgue sense and give rise to important topological invariants [1, 7, 17].

In general QFTs, however, graph integrals require renormalization, and different procedures may produce different values. Our work asks whether the geometry of holomorphic field theories on Kähler manifolds permits a canonical construction.

For a graph Γ with vertex set Γ_0 and edge set Γ_1 , assign a point $x_v \in M$ to each vertex. Each edge $e = (v, w)$ contributes a propagator $P_e(x_v, x_w)$, constructed from a Green kernel and its derivatives. The graph integral has the schematic form

$$\int_{M^{\Gamma_0}} \prod_{e=(v,w) \in \Gamma_1} P_e(x_v, x_w),$$

where $M^{\Gamma_0} = \prod_{v \in \Gamma_0} M$. The propagators are singular when their endpoints coincide (i.e., $x_v \rightarrow x_w$). The main analytic difficulty is to control simultaneous and nested collisions of vertices.

In joint work with Wang, we proved the convergence of these graph integrals on closed real-analytic Kähler manifolds [22]. Using heat-kernel techniques together with Getzler rescaling from local index theory, we show that the graph integrands extend canonically to the Fulton–MacPherson compactification of configuration spaces as forms with controlled singularities and therefore admit well-defined Cauchy principal-value regularizations.

When specialized to Calabi–Yau threefolds, this gives a direct and mathematically rigorous construction of candidate higher-genus ($g \geq 2$) B-model amplitudes through the Feynman graph integrals proposed by BCOV in [2]. In complex dimension one, earlier work of Li [18] and Costello–Li [9] constructed a punctured version of BCOV theory for elliptic curves whose large-complex-structure limit recovers punctured Gromov–Witten invariants of the mirror. This provides an important rigorous higher-genus instance of the mirror-symmetry picture underlying the BCOV construction.

A further question is whether different natural renormalization procedures produce the same invariant. With Wang, we proved that heat-kernel renormalization, zeta regularization, and Cauchy principal-value regularization all give the same value for holomorphic field theories on closed real-analytic Kähler manifolds [23]. The difficulty is that these procedures organize the same singularities in very different ways: heat-kernel renormalization cuts off short time, zeta regularization uses meromorphic continuation, and principal value uses the geometry of the resolved diagonals. Wonderful compactifications in algebraic geometry provide a common geometric framework for analyzing the nested singularity loci underlying all three regularizations, allowing the three procedures to be compared directly.

We also derived a gauge anomaly formula: the failure of the resulting graph currents to be $\bar{\partial}$ -closed is expressed by boundary integrals over the compactified space of heat-kernel parameters [23, §6]. In the Batalin–Vilkovisky formalism, the short-time boundary contributions compute the obstruction to the quantum master equation.

We will next extend the construction to families and study anomaly formulas for Feynman graph integrals under variations of Kähler and complex structures. The goal is to understand the differential forms on moduli spaces of Calabi–Yau manifolds obtained from fiberwise Feynman graph integrals, in analogy with the analytic torsion forms discussed in §3. The main analytic problem is to obtain estimates on the propagators and their singularities that are uniform in the family parameter. A second, longer-term program is field theory on manifolds with boundary, where my Witten-deformation gluing methods may be used to separate bulk and boundary contributions.

3 Higher Cheeger–Müller/Bismut–Zhang theory

A classical problem in global analysis is to understand why spectral invariants of differential operators can coincide with invariants defined by topological or combinatorial methods. Ray–Singer analytic torsion is a central example: it is defined from regularized determinants of Hodge Laplacians, whereas

Reidemeister torsion is constructed from a finite-dimensional cellular complex. The Cheeger–Müller theorem identifies analytic and Reidemeister torsion, and Bismut–Zhang extended the comparison to a more general setting using Witten deformation [6].

For families of manifolds, the corresponding problem is to compare differential-form-valued analytic torsion with higher topological torsion over the parameter space. Let $\pi : M \rightarrow S$ be a smooth fibration with fibers $Z_s = \pi^{-1}(s)$. Bismut and Lott generalized the numerical analytic torsion of a single manifold to an even differential form for a flat complex vector bundle $F \rightarrow M$

$$\mathcal{T}_{\text{BL}}(M/S, F)$$

on the parameter space S [5]. Its degree-zero component recovers ordinary analytic torsion. On the topological side, higher Franz–Reidemeister, or Igusa–Klein, torsion can detect exotic smooth structures that are invisible to ordinary characteristic classes [13–15].

A higher Cheeger–Müller/Bismut–Zhang theorem should compare these analytic and topological classes. The difficulty is that earlier approaches remove different restrictions. The analytic method of Bismut–Goette [3] treats general flat bundles but requires the existence of a fiberwise Morse function satisfying fiberwise Thom–Smale transversality. On the other hand, the axiomatization approach in [20] avoids the fiberwise Morse-function existence assumption but works only for trivial flat bundles.

With Martin Puchol, we combine the advantages of these approaches. The axiomatization argument reduces the problem to the comparison of relative higher torsions. On the analytic side, the relative torsion is

$$\mathcal{T}_{\text{BL}}^{\text{rel}}(M/S, F) := \mathcal{T}_{\text{BL}}(M/S, F) - \mathcal{T}_{\text{BL}}(M/S, \underline{\mathbb{C}}^{\text{rk}(F)}),$$

where $\underline{\mathbb{C}}^{\text{rk}(F)}$ is the trivial flat bundle of the same rank as F ; the topological torsion is normalized in the same way.

We extend the analytic method of Bismut–Goette from fiberwise Morse functions to fiberwise generalized Morse functions, which always exist but are allowed to have birth–death singularities. This simultaneously handles general unitary flat bundles and removes the fiberwise Morse-function existence restriction, yielding the expected higher Cheeger–Müller/Bismut–Zhang comparison for the relative torsions, while still assuming Thom–Smale transversality [19].

The main analytic difficulty is that the classical Witten-deformation estimates cease to be uniform near a birth–death singularity. At a birth–death point, two critical points merge and the Hessian develops a zero direction, so the standard quadratic Morse model degenerates. We first isolate the birth–death points using a localization technique related to the non-Morse Witten deformation I developed for torsion gluing formulas [24, 25]. The remaining issues can then be resolved by combining Bismut–Lebeau’s localization techniques [4] with the noncompact Agmon estimates from my earlier joint work with Dai [10].

Future direction: removing transversality by higher structures

The remaining step toward a fully general theorem is to remove global Thom–Smale transversality. Since a single fiberwise metric satisfying this condition need not exist globally, my approach is to choose Thom–Smale metrics locally on an open cover of the base and glue the resulting Morse complexes by higher continuation data.

One can cover the base by sufficiently small open sets $S = \bigcup_i U_i$ such that, over each U_i , there exists a smooth family of metrics in the vertical direction satisfying the Thom–Smale condition. These local Thom–Smale complexes do not agree literally on overlaps. On a double intersection $U_i \cap U_j$, two choices of metric give a continuation map between the corresponding complexes. On a triple intersection $U_i \cap U_j \cap U_k$, the two possible compositions of continuation maps agree only up to chain homotopy. On higher intersections, one must introduce higher homotopies satisfying suitable coherence relations.

This higher structure should define a Čech version of the relative topological torsion, while local Witten deformation should identify it with the Bismut–Lott analytic torsion form. The new analytic issue is to control continuation maps and higher homotopies uniformly near birth–death loci, extending the estimates used in our comparison theorem.

4 Analysis of Schrödinger-type operators on noncompact manifolds

A basic problem in spectral geometry is to understand spectral invariants of Schrödinger-type operators on noncompact manifolds. This question also arises naturally in the Landau–Ginzburg B-model. An LG model consists of such a manifold X and a holomorphic function f , called the superpotential. The associated operator on differential forms has the form

$$\Delta_f = \Delta + |\nabla f|^2 + \text{a zero-order term involving } \text{Hess}(f).$$

Regularized determinants of this operator are expected to govern the genus-one theory.

A major obstacle in this program is that many fundamental tools of spectral geometry rely heavily on compactness and do not directly extend to noncompact manifolds with potentials. In a series of joint works with X. Dai, we developed analytic foundations for Schrödinger-type operators on noncompact manifolds, which are needed to define higher-genus BCOV invariants in the LG B-model. We proved exponential decay of eigenfunctions near infinity (Agmon estimates), which leads to the well-definedness of Thom–Smale–Witten cohomology for Morse functions on noncompact manifolds [10]. We also derived heat-kernel asymptotic expansions for such operators [11], allowing us to extend the local index theorem to noncompact manifolds with potentials.

Weyl law for Schrödinger operators on noncompact manifolds

A basic question in spectral geometry is whether the classical Weyl law continues to hold for Schrödinger operators on noncompact manifolds, where both the geometry at infinity and the growth of the potential can affect the eigenvalue distribution.

Using the heat-kernel expansion developed in [11], we established in [12] a *non-semiclassical Weyl law* for Schrödinger operators on noncompact manifolds with bounded geometry and potentials satisfying a doubling condition. This gives a Weyl law in a broad noncompact geometric setting without prescribing a specific model for the geometry at infinity. Even in special cases, such as hyperbolic manifolds with rapidly growing potentials studied in [16], which fall within the scope of our result, the analysis is already delicate.

In subsequent work with Braverman and Dai, we removed both the bounded geometry and doubling assumptions [8]. Our intrinsic criterion balances local curvature and injectivity radius against the growth and oscillation of the potential. It allows geometric irregularity at infinity, provided that this balance is preserved. We also constructed examples in which the Weyl law fails when the balance breaks down.

More recently, I applied these ideas to show that the weighted Laplacian on every complete gradient Ricci shrinker satisfies the classical Weyl law [26]. The difficulty is that, although the criterion in [8] is very general, it still cannot be applied directly to Ricci shrinkers. The key observation is that, roughly speaking, the problematic locus where this criterion may fail is asymptotically negligible. To establish this, I prove a large-scale regularity estimate showing that, in geodesic balls of radius R , the region with curvature radius below R^{-1} occupies a vanishing proportion of the volume as $R \rightarrow \infty$.

My next goal is to go beyond the leading Weyl asymptotics and obtain quantitative remainder estimates. On compact manifolds, refined estimates are derived from the microlocal structure of the wave kernel and Fourier Tauberian arguments, and are closely related to the dynamics of the geodesic flow. Extending these techniques to Schrödinger operators on noncompact manifolds would require controlling simultaneously the Hamiltonian dynamics and the geometry at infinity.

5 Toward a higher-genus Calabi–Yau/Landau–Ginzburg correspondence on the B-side

My long-term goal is a higher-genus Calabi–Yau/Landau–Ginzburg (CY/LG) correspondence on the B-model side, relating a compact Calabi–Yau theory to a noncompact LG model. The theory is organized by genus. At genus zero, the B-model is governed by tt^* -geometry, which can be viewed as a generalized variation of Hodge structures. With Xinling Tang, we proved the CY/LG correspondence for Weil–Petersson metrics and tt^* -structures [21].

Genus one. My immediate goal is a genus-one comparison for Calabi–Yau threefolds. On the LG side, I am studying a suitably corrected analytic torsion built from regularized determinants of the deformed Laplacian. In ongoing work, I have derived its anomaly formula under variations of the superpotential and compared it with the anomaly formula for BCOV torsion on Calabi–Yau threefolds. This work is currently being prepared for publication. Agreement of the anomaly formulas would not by itself identify the invariants: a further task is to determine the remaining ambiguity through normalization and moduli boundary asymptotics. Our genus-zero identification of Weil–Petersson metrics and tt^* -structures provides a natural starting point for this comparison.

Higher genus. For $g \geq 2$, the next problem is to combine the graph-integral construction of §2 with the noncompact spectral analysis of §4. On the compact CY side, our graph integrals define candidate higher-genus B-model invariants. On the LG side, a new difficulty arises because vertices can collide while simultaneously escaping to infinity. Estimates near collision diagonals and estimates at infinity must therefore remain compatible when both phenomena occur together.

My approach is to construct a compactification of the configuration space that records both collision diagonals and escape to infinity, and to establish propagator estimates that remain uniform near their intersections. The compactification should distinguish the two different degeneration patterns so that their analytic behavior can be studied. The main task is to determine the asymptotics of the propagators near each boundary stratum and prove that the resulting graph integrals admit a canonical regularization.

This would provide a rigorous construction of candidate higher-genus B-model invariants on the LG side, making their comparison with the compact CY construction a precise mathematical problem.

Together with the genus-one torsion comparison, this program brings together the main techniques developed in my work: configuration-space compactifications for collision singularities, heat-kernel and spectral estimates for noncompact Schrödinger operators, and localization methods associated with the superpotential. The ultimate goal is to construct the higher-genus invariants on both sides within compatible analytic frameworks and prove their correspondence.

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